

**Astronomy C - Rickards Astronomy 2025 - Rickards Invitational Div. C SATELLITE**

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**Instructions**

You have 50 minutes to take this exam once you begin. As per the rules, you are allowed to use either physical or digital notes, as well as calculators for the exam. Out of browser time will not be tracked, but using the internet/generative AI is NOT allowed at any point in the exam. Any discovered violations will receive no points for that answer, and repeated incidents will result in disqualification.



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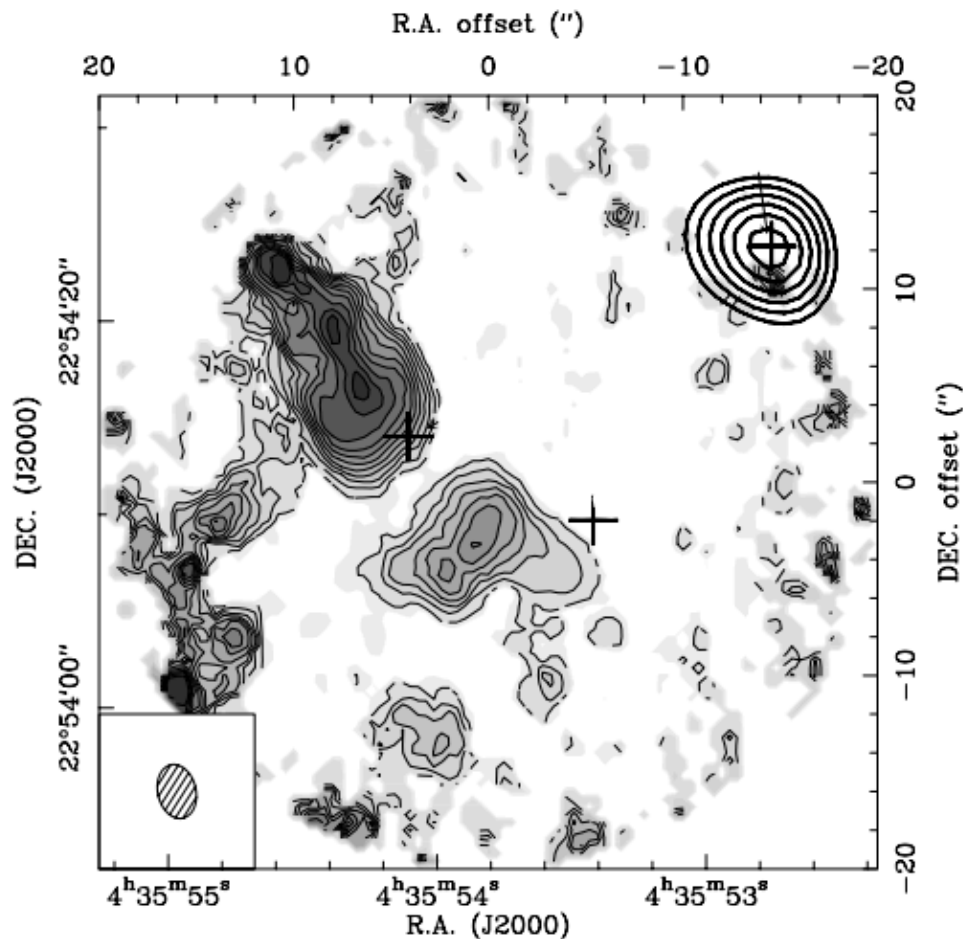
**Section 1: DSOs**

Use the following image to answer question 1:

**1. (7.00 pts)**

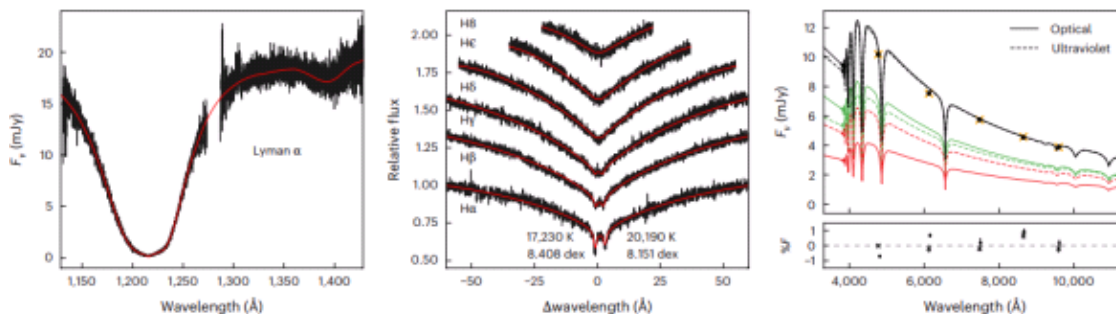
- What is the name of this DSO? (1)
- Which observatory was this image taken at? (1)
- The bright red region in the middle left of the image is a(n) \_\_\_\_\_, and composed primarily of \_\_\_\_\_. (2)
- Would you expect the gas in this region to be primarily ionized or neutral? Why? (3)

Use the following image to answer question 2:



2. (8.00 pts)
- What is the name of the DSO shown in this image? (1)
  - This diagram shows the presence of which compound? Why is it significant? (4)
  - What type of star is this DSO? Be specific! (2)
  - True/False: This DSO is a type of pulsating variable star. (1)

Use the following image to answer question 3:

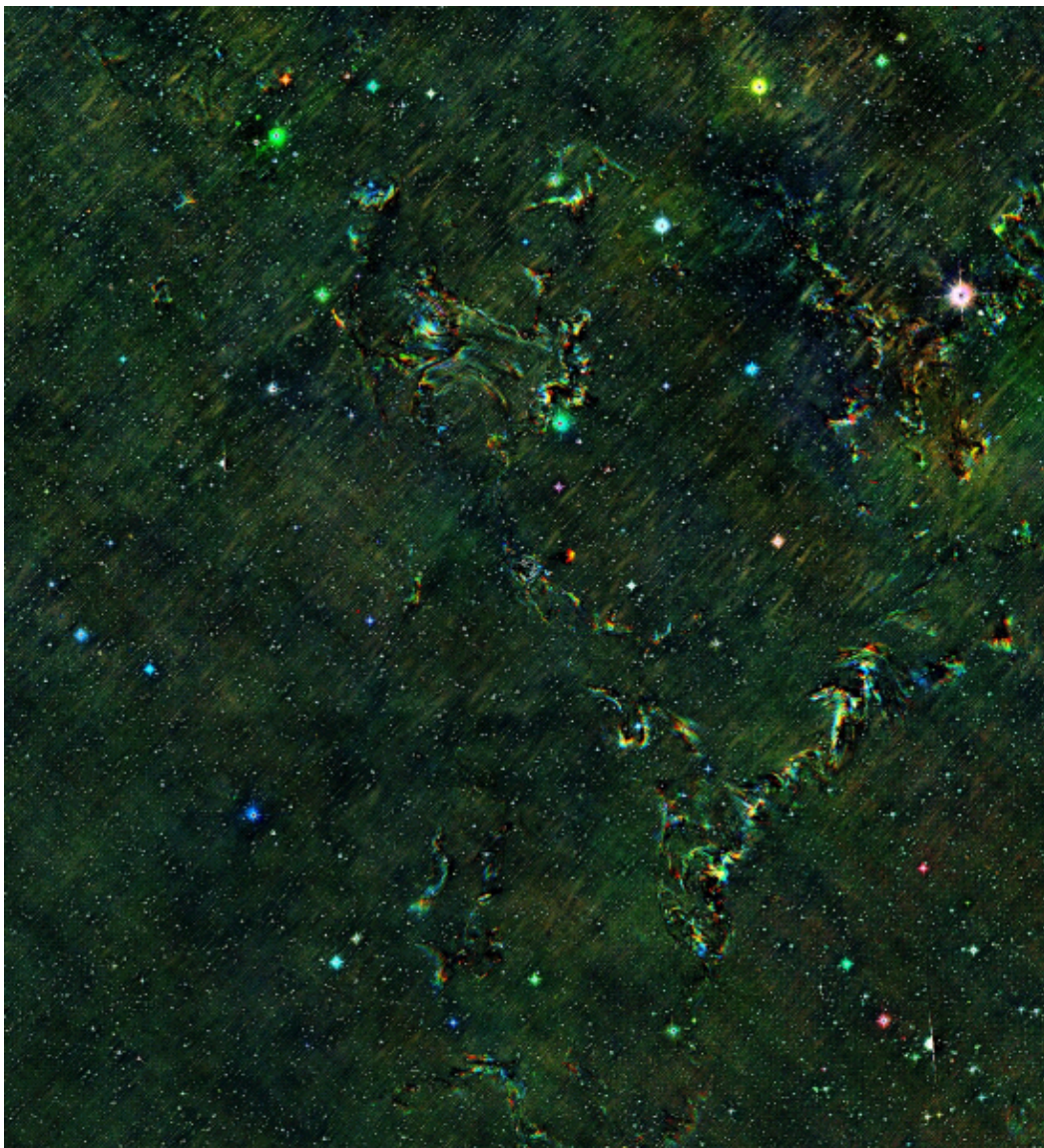




**3. (8.00 pts)**

- These graphs represent which DSO? (1)
- What is unique about this particular DSO? (2)
- This DSO “violates” a well-known astrophysics principle. What is this principle and how is a violation possible? (4)
- When was this DSO first detected? (1)

Use the following image to answer question 4:



**4. (9.00 pts)**

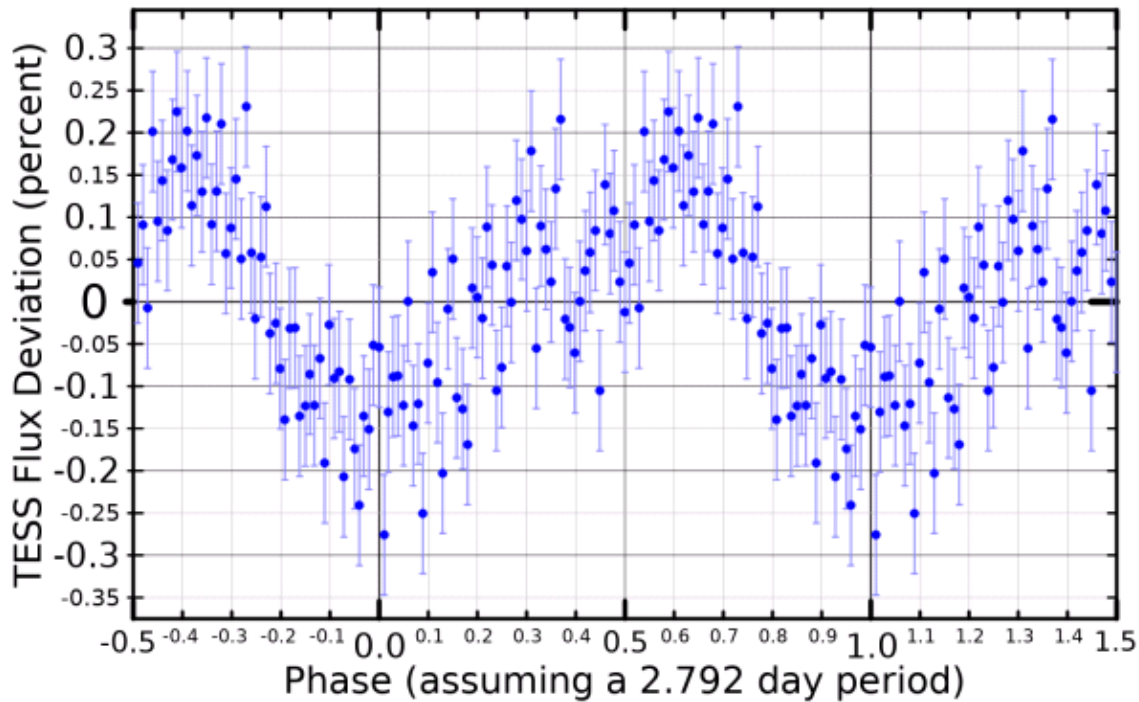
- a. What is the name of the DSO shown in the image? (1)
- b. What section of the EM spectrum was this image taken in? What is this image showing (3)?
- c. True/False: this image is a composite taken over the course of 5 years. (1)
- d. This DSO had an unusual element detected in it. Which element was this and why was that significant? (3)
- e. This DSO emits radiation most prominently in \_\_\_\_\_ wavelengths. (1)

Use the following image to answer question 5:

**5. (8.00 pts)**

- a. Provide 2 names for the DSO shown in the image. (2)
- b. This DSO is primarily observed in which wavelengths? These wavelengths correspond to which colors in the image? (2)
- c. This DSO has recently been observed to be interacting with another object. What is this object? (official designations are not needed) (2)
- d. This image uses data from two different telescopes. What are they? (2)

Use the following image to answer question 6:



**6. (6.00 pts)**

- What is the DSO shown by the light curve in the image? (1)
- This DSO is the first of its kind found to contain what structures? Briefly describe them and name the most probable mechanism of their formation. (3)
- True/False: this DSO is confirmed to have two planets orbiting it. (1)
- This DSO is relatively famous, with a popularized (if dramatic) name being given to it in pop culture. What is this name? (1)

## Section 2: Theory

**7. (3.00 pts)**

What kind of variable stars are RR Lyrae variables classified as? What is the name of the leading hypothesis of RR Lyrae periodicity?

**8. (4.00 pts)**

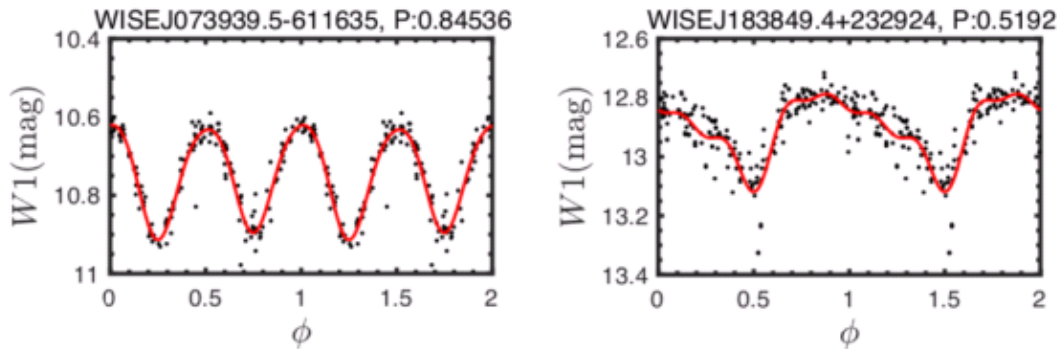
Which phenomenon is responsible for the Instability Strip on the HR diagram (and many other types of variability in stars)? Briefly explain how it works.

**9. (2.00 pts)**

Assume you are observing a variable star which is very red in color, has a pulsation period of about 125 days, and is a late-stage giant. What type of variable is this?

**10. (4.00 pts)**

This image shows two types of variable stars. Which types of variable stars do these graphs show? Be sure to note which one is which in your answer.



**11. (2.00 pts)** Are Wolf-Rayet stars classified as Population I or II? Explain.

**12. (1.00 pts)** Globular clusters are mostly composed of Population I stars.

A) True B) False

**13. (5.00 pts)**

What technique is used to dissect signals into their constituent frequencies? What other uses does it have in radio astronomy?

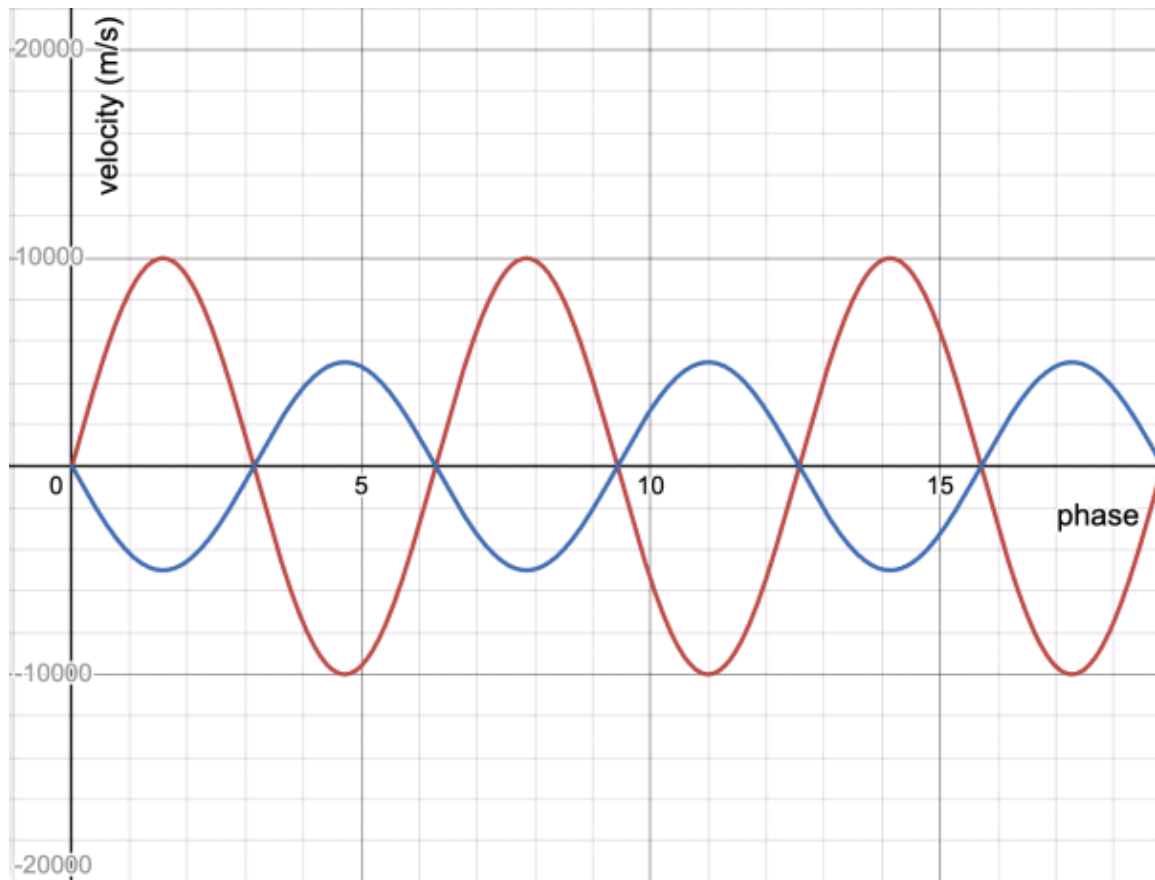
**14. (2.00 pts)** Which wavelengths would be best used to image a neutron star?

**15. (1.00 pts)** Most Type II supernovae do not occur in elliptical galaxies.

A) True B) False

## Section 3: Astrophysics



**16. (36.00 pts)**

This is the radial velocity graph of a system of two white dwarves.

- Given that the total mass of the system is  $1.5 M_{\odot}$ , what is the period of the white dwarves? (4)
- What is the total orbital energy of this system? (2)
- Due to general relativity, this energy is lost to gravitational radiation at a rate of

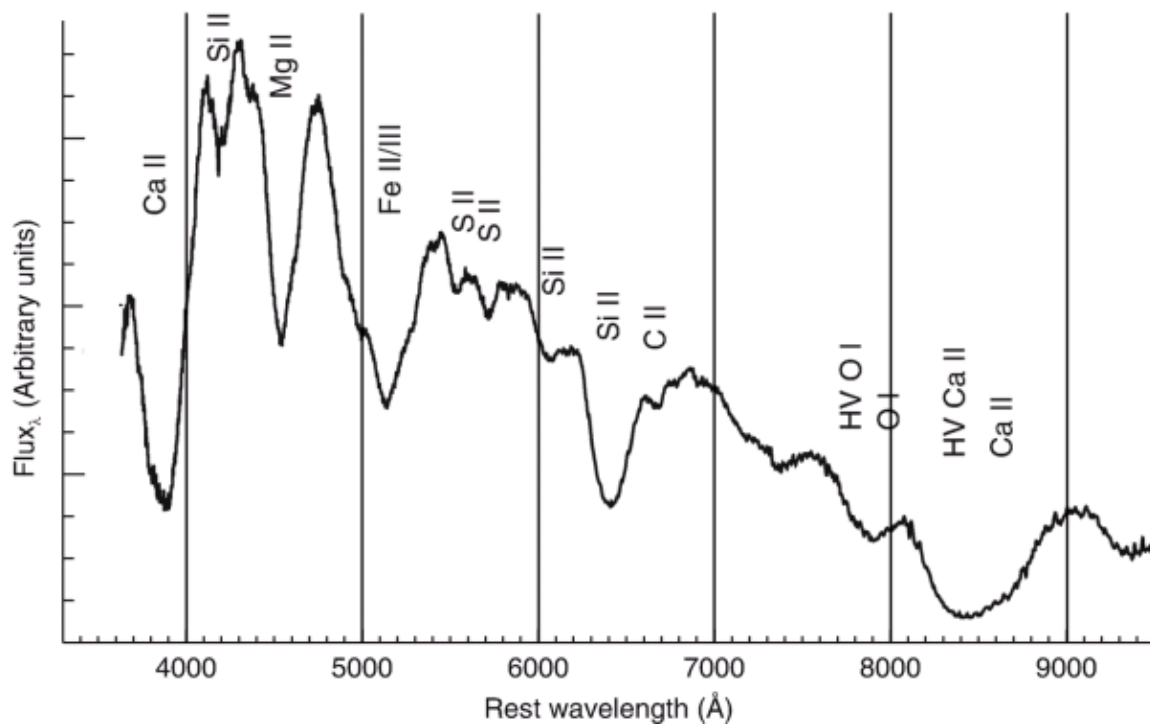
$$-\frac{32}{5} G^4 c^{-5} r^{-5} (M_a M_b)^2 (M_a + M_b).$$

What is the rate of orbital decay of the two white dwarfs? (5)

- How long will it take these two white dwarves to merge? (6)
- What type of supernova will this collision result in? (1)

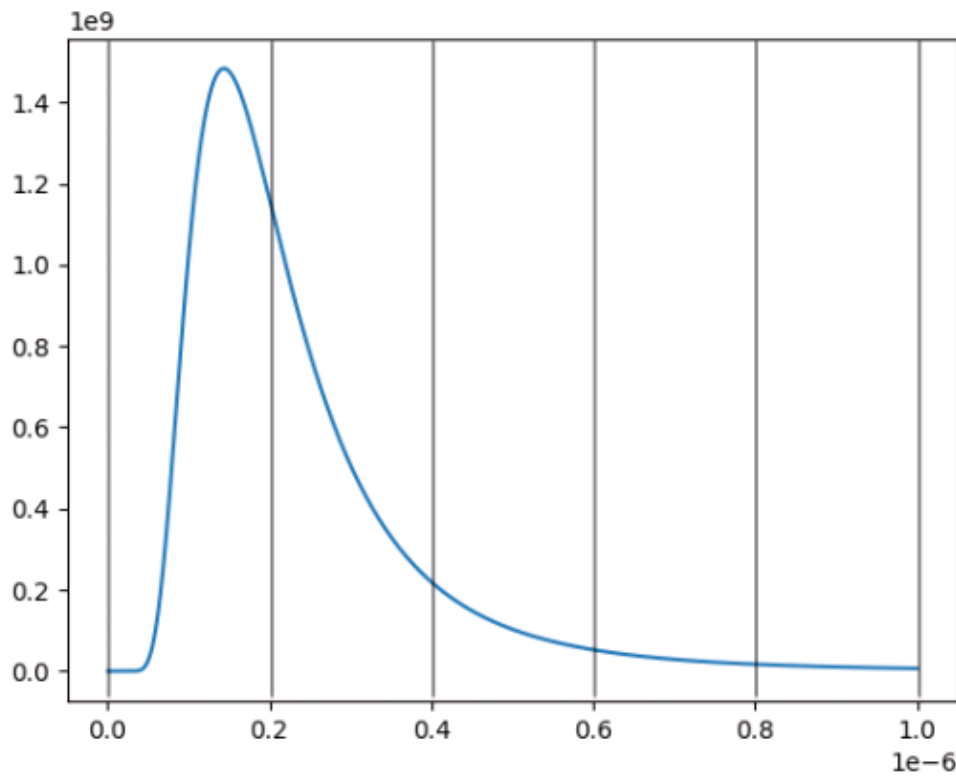
**17. (12.00 pts)**





- This will be the spectrum of the supernova. How fast is the system moving from Earth? (4)
- The parallax of the white dwarves is  $0.005''$ . Is the value from (a.) reasonable? (2)

**18. (46.00 pts)**



This is the current light spectrum of the system referenced in #16 and #17.

- What is the temperature of the white dwarfs? (2)
- Light is absorbed as it passes through the ISM through a process known as extinction. In the milky way galaxy, extinction is about 1.8 magnitudes/kpc. What is the apparent magnitude of the system? (Use the nonrelativistic equations for this problem). (6)
- A planet exists in this system. How would it orbit the white dwarves? (2)
- The transit depth if all three members of the system were to line up with the planet in front is 0.6. Provide a simplified expression for the inclination of the planet in terms of its orbital distance. (5)
- What will be the temperature of the white dwarfs at the time of collision? You may use  $t=10^{30}$ s if you did not obtain a value for merger time. (8)